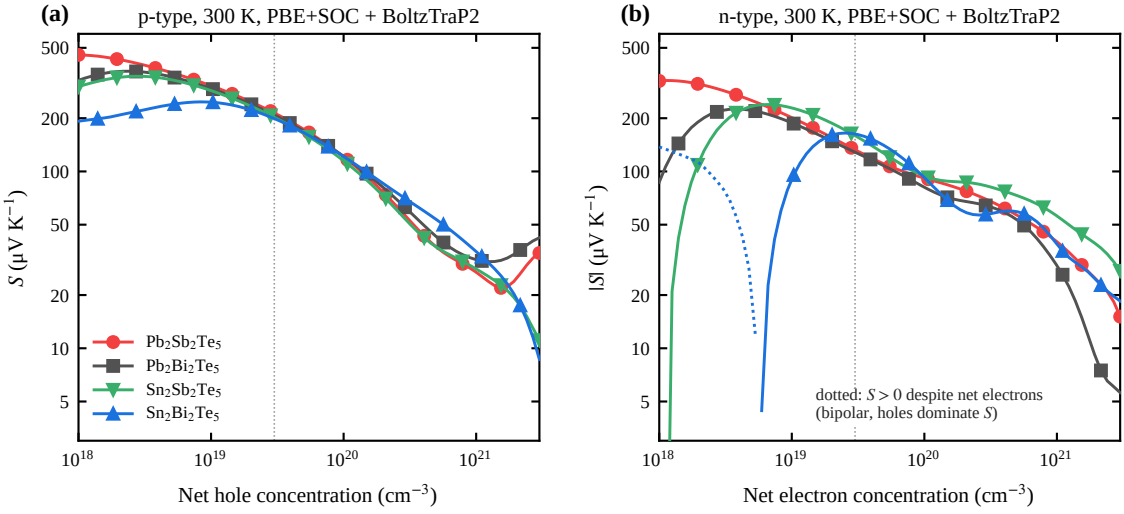
(a)
p-type, 300 K, PBE+SOC + BoltzTraP2
S (μV K−1)
500
200
100
50
20
10
5
Pb2Sb2Te5
Pb2Bi2Te5
Sn2Sb2Te5
Sn2Bi2Te5
10^18
10^19
10^20
10^21
Net hole concentration (cm−3)
(b)
n-type, 300 K, PBE+SOC + BoltzTraP2
|S| (μV K−1)
dotted: S > 0 despite net electrons
(bipolar, holes dominate S)
Net electron concentration (cm−3)